

Volume Integrals

Q. If $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$, then evaluate

$\iiint_V \nabla \cdot \vec{F} \, dv$, when V is bounded by the planes

$$x=0, y=0, z=0 \text{ and } 2x+2y+z=4$$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2x^2 - 3z) + \frac{\partial}{\partial y}(-2xy) + \frac{\partial}{\partial z}(-4x)$$

$$= 4x - 2x = 2x$$

$$\therefore \iiint_V \nabla \cdot \vec{F} \, dv = \iiint_V 2x \, dx \, dy \, dz$$

$$= \int_0^2 \int_0^{2-x} \int_0^{4-2x-2y} 2x \, dz \, dy \, dx$$

$$= 2 \int_0^2 \int_0^{2-x} xz \Big|_0^{4-2x-2y} dy \, dx$$

$$= 2 \int_0^2 \int_0^{2-x} x(4-2x-2y) dy \, dx$$

$$= 2 \int_0^2 \int_0^{2-x} (4x - 2x^2 - 2xy) dy \, dx$$

$$= 2 \int_0^2 \left[4xy - 2x^2y - y^2x \right]_0^{2-x} dx$$

$$= 2 \int_0^2 (4x(2-x) - 2x^2(2-x) - (2-x)^2x) dx$$

$$= 2 \int_0^2 (8x - 4x^2 - 4x^2 + 2x^3 - (4x + 4x^2 - x^3)) dx$$

$$+ 2 \int_0^2 (4x - 4x^2 + x^3) dx$$

$$= 2 \left[4x^2 - \frac{4}{3}x^3 + \frac{x^4}{4} \right]_0^2$$

$$= 2 \left[2 \cdot 4 - \frac{4}{3} \cdot 8 + 4 \right]$$

$$= 2 \left[8 - \frac{32}{3} + 4 \right]$$

$$= 2 \left[12 - \frac{32}{3} \right] = 2 \left(\frac{36 - 32}{3} \right)$$

$$= \frac{8}{3}$$

Q. If $\vec{A} = 2xz \hat{i} - x\hat{j} + y^2 \hat{k}$, evaluate $\iiint_V \vec{A} dV$, where V is the region bounded by the surface $x=0, y=0, x=2, y=6, z=x^2, z=4$?

$$\iiint_V \vec{A} dV = \iiint_V (2xz \hat{i} - x\hat{j} + y^2 \hat{k}) dx dy dz$$

$$= \int_0^2 \int_0^6 \int_{x^2}^4 (2xz \hat{i} - x\hat{j} + y^2 \hat{k}) dz dy dx$$

$$= \hat{i} \int_0^2 \int_0^6 \int_{x^2}^4 2xz dz dy dx - \hat{j} \int_0^2 \int_0^6 \int_{x^2}^4 x dz dy dx + \hat{k} \int_0^2 \int_0^6 \int_{x^2}^4 y^2 dz dy dx$$

$$= \hat{i} \int_0^2 \int_0^6 (xz^2)_{x^2}^4 dy dx - \hat{j} \int_0^2 \int_0^6 xz \Big|_{x^2}^4 dy dx + \hat{k} \int_0^2 \int_0^6 y^2 z \Big|_{x^2}^4 dy dx$$

$$= \hat{i} \int_0^2 \int_0^6 x(16 - x^4) dy dx - \hat{j} \int_0^2 \int_0^6 x(4 - x^2) dy dx + \hat{k} \int_0^2 \int_0^6 y^2(4 - x^2) dy dx$$

$$= i \int_0^2 (16x - x^5) (y)_0^6 dx - j \int_0^2 (4x - x^3) (y)_0^6 dx +$$

$$K \int_0^2 (4 - x^2) (y_{3/3})_0^6 dx$$

$$= i \int_0^2 6(16x - x^5) dx - j \int_0^2 6(4x - x^3) dx + K \int_0^2 72(4 - x^2) dx$$

$$= 6i \left[8x^2 - \frac{x^6}{6} \right]_0^2 - 6j \left[2x^2 - \frac{x^4}{4} \right]_0^2 + 72K \left[4x - \frac{x^3}{3} \right]_0^2$$

$$= 6i \left[32 - \frac{64}{6} \right] - 6j \left[8 - 4 \right] + 72K \left[8 - \frac{8}{3} \right]$$

$$= (192 - 64)i - 6 \times 4j + 72 \times \frac{(24 - 8)}{3} K$$

$$= 128i - 24j + 24 \times 16 K$$

$$= 128i - 24j + 384K$$

$$\begin{array}{r} 4 \times 4 \times 4 \\ 16 \times 2 \\ 32 \end{array}$$

$$\begin{array}{r} 24 \times 2 \\ 16 \times 2 \\ 24 \\ 384 \end{array}$$

$$\sum_{i=1}^n (x_i^2 + y_i^2 + z_i^2) = \sum_{i=1}^n (x_i^2 + y_i^2 + z_i^2)$$

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$$\sum_{i=1}^n (x_i^2 + y_i^2 + z_i^2) = \sum_{i=1}^n (x_i^2 + y_i^2 + z_i^2)$$